Hawai‘i – Hurricane ESF-6 playbook

Data as of 2026-10-27

County: Hawai‘i · **Hazard:** Hurricane

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **202,163**	Shelter demand (central) **6,065**	Shelter demand low – high **4,043 – 10,108**
Limited English (C) **5,371**	Power-dependent (M) **1,450**	Disability (I) **30,614**
Age 65+ (I) **44,910**	People without a vehicle (T) **9,876**	Known 2026 sites usable **10**

Shelter-seeking rates: low 2%, central 3%, high 5.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

County total = upper bound. It assumes every at-risk tract is hit at once (all tracts). Real events usually hit fewer communities.

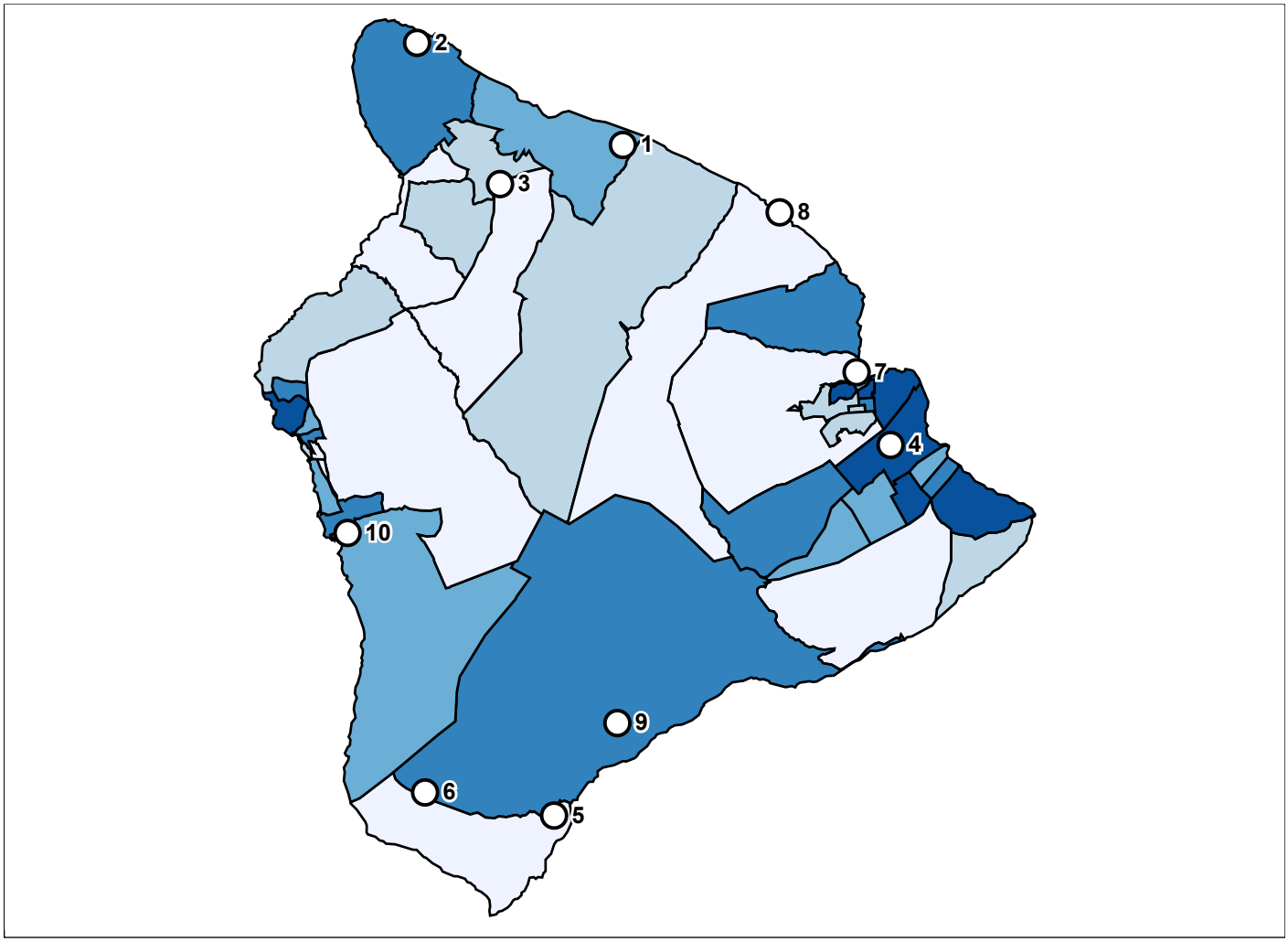
Largest single community (highest-demand census tracts)

Hurricanes affect the whole county, so these are simply the most populous at-risk tracts.

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
Tract 210.14 · Hawaiian Paradise Park	8,135	163 / **244** / 407	260	49	834	1,504	0
Tract 213 · Honaunau-Napoopoo	7,384	148 / **222** / 369	176	66	777	2,042	170
Tract 208.02 · Hilo	7,010	140 / **210** / 350	50	64	1,087	1,974	358

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–715 people (class 1 of 5)
- AFN need 715–1,017 people (class 2 of 5)
- AFN need 1,017–1,189 people (class 3 of 5)
- AFN need 1,189–1,430 people (class 4 of 5)
- AFN need 1,430–2,219 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Honokaʻa Gym	45-541 Lehua St., Honokaʻa	no site-level screen for this hazard	1
2	✓ **Usable**	Ikuo Hisaoka Gym	54-382 Kamehameha Park Road, Kapaʻau	no site-level screen for this hazard	1
3	✓ **Usable**	Kahilu Town Hall	67-1182 Lindsey Road, Waimea	no site-level screen for this hazard	1
4	✓ **Usable**	Keaʻau Armory	16-512 Volcano Road, Keaʻau	no site-level screen for this hazard	1
5	✓ **Usable**	Nāʻālehu Elementary School Gym	95-5545 Mamalahoa Highway, Nāʻālehu	no site-level screen for this hazard	1
6	✓ **Usable**	Ocean View Community Center	92-8924 Leilani Circle, Ocean View	no site-level screen for this hazard	1
7	✓ **Usable**	Puʻuʻeo Community Center	145 Wainaku St., Hilo	no site-level screen for this hazard	1
8	✓ **Usable**	Pāpaʻaloa Park Annex	35-1994 Government Main Road, Pāpaʻaloa	no site-level screen for this hazard	1
9	✓ **Usable**	Robert N. Herkes Gym	96-1219 Kamani St., Pāhala	no site-level screen for this hazard	1
10	✓ **Usable**	Yano Hall	82-6156 Māmalahoa Highway, Captain Cook	no site-level screen for this hazard	1

Candidate public schools in this county usable for this hazard: **56**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	6,065	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	121,298	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	242,596	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	6,065	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	12,130	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	152	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	304	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	410	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	44	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	9,876	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	970	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	566	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	1,291	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	41	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Other English-based Creole languages	25	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	17	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	15	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	11	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Chinese	8	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	8	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	6	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Central Pacific hurricane season start (June 1) NOAA Central Pacific Hurricane Center	Confirm partner capacity pledges (cots, meals, generators, accessible vehicles, interpreters) and the list of sites usable for wind; refresh priority-outreach tracts.	County Emergency Management (each county)	dhs, hiema, red_cross
BEFORE	Hurricane Watch NOAA NWS / Central Pacific Hurricane Center	Contact priority tracts (power-dependent, no-vehicle, limited English); confirm which shelters will open, generator power for medical devices, and accessible transport.	County Emergency Management (each county)	doh, utilities, dot, dhs
BEFORE	Hurricane Warning NOAA NWS / Central Pacific Hurricane Center	Open shelters including AFN-capable sites; start accessible transport for residents without vehicles; send pre-approved multilingual messages.	County Emergency Management (each county)	red_cross, dhs, doe, dot
AFTER	Warning cancelled or all clear County / NWS / PTWC	Feeding, re-power for medical devices, recovery sheltering for displaced households; coordinate donations and volunteers.	American Red Cross	hawaii_voad, salvation_army, faith_based, utilities

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	Tract 205 · Hilo	5,552	31	836	149	1,203	2,219
2	Tract 211.08 · Hawaiian Beaches / Nanawale Estates	6,155	43	1,497	56	486	2,082
3	Tract 210.03 · Ainaloa / Orchidlands Estates	7,009	35	1,593	101	259	1,989
4	Tract 206 · Hilo	6,802	45	1,261	67	387	1,760
5	Tract 203 · Hilo	4,167	32	670	31	991	1,725
6	Tract 208.02 · Hilo	7,010	64	1,087	50	358	1,558
7	Tract 215.04 · Kailua	6,416	28	801	263	463	1,555
8	Tract 210.13 · Kurtistown	5,059	41	955	207	286	1,488
9	Tract 218 · Hawi / Kapaau	6,435	60	1,134	182	55	1,430
10	Tract 215.12 · Kaiminani	4,594	29	578	77	646	1,330

Languages: people who speak English less than "very well" (Hawaii)

Language	People 5+	90% MOE
Ilocano	3,409	±869
Other English-based Creole languages	2,040	±740
Japanese	1,408	±504
Spanish	1,217	±499
Tagalog, Filipino	924	±368
Chinese	651	±326
Hawaiian	637	±264
Marshallese	538	±303

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
2020 Census Block Assignment File, place/CDP (tract community labels)	2026-10-27	https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip
2020 Census PL 94-171 block population P1_001N (tract community labels)	2026-10-27	https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.